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IT FDN 110 GP: Foundations of Python Programming
Assignment05
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Advanced Collections and Error Handling

Introduction

Assignment05 demonstrated the use of constants, variables, print statements, data processing using dictionaries, and exception handling. I created a console program that registers students for courses, shows the current registrations, and saves them to a JSON file. There are four options on the menu where a user can: (1) register a student for a course, (2) show the current data, (3) save the data to a file, and (4) exit the program.

Data Collected

This program involved importing the Python standard library, json.

Each registration stores three values: student's first name, last name, and course name. The program keeps every registration in memory as a list of dictionaries.

Constants

- MENU (str): the menu text with options 1 to 4
- FILE_NAME (str): "Enrollments.json"

Variables

- student_first_name: str is set to an empty string '' and is the first name typed by the user
- student_second_name: str is set to an empty string '' and is the last name typed by the user
- course_name: str is set to an empty string '' and is the course name typed by the user
- menu_choice: str is set to an empty string ''
- student_data: is set to an empty dict {} and holds the student's first name, last name and the course
- students: list is set to an empty list [] and for every registration
- file: set to None and the open file while reading or writing

How the data is organized

The dictionary in this program has three keys: FirstName, LastName, and CourseName

The file, Enrollments.json, contains the list as a JSON array, for example:

[{"FirstName": "Bugs", "LastName": "Bunny", "CourseName": "Python 100"}]

The program

When the program starts, it imports Python's built-in json module and also defines constants and variables. It opens Enrollments.json in read mode and loads it with json.load() function. Then it closes the file, and if there's an error, it prints a message (see Error handling below).

The menu uses a "while True" loop that prints a message, "What would you like to do?" and runs the appropriate part of the program based on the selection.

- Choice 1 prompts the user to input the first name, last name, and course name with the input() function. It builds student_data dictionary from the 3 answers, appends it to students, and prints a confirmation message.
- Choice 2 prints current data in Enrollments.json.
- Choice 3 writes the data with json.dump(), closes the file, and a confirmation message is printed: "The data has been saved in FILE_NAME".
- Choice 4 exits the loop with break and prints the message, "Program ended. Have a nice day!"
- I used an "else:" for any invalid choices other than 1 to 4, which results in the printed statement, "Please only choose option 1, 2, 3, or 4."

Error handling

The program uses three try/except blocks so that a bad file or bad input prints a message to keep the program running.

- FileNotFoundError (at the start if Enrollments.json is missing) prints a message, "Text file must exist before running this script," and displays a technical error message.
- Exception (at the start for any other read problem) prints a message, "There was a non-specific error!" and displays a technical error message.
- ValueError (at option 1 for first name) uses .isalpha() function and prints a message, "The first name should not contain numbers."

- ValueError (at option 1 for last name) uses .isalpha() function and prints a message, “The last name should not contain numbers.”
- Exception (at option 1 at input for a non-specific error) and prints a message, “There was a non-specific error!” and displays a technical error message.
- TypeError (at option 3 for data that can’t be converted to JSON) and prints a message, “Please check that the data is a valid JSON format,” and displays a technical error message.
- Exception (at option 3 for any other write problem) and prints a technical error message.

Below are screenshots of the program and testing I did in PyCharm, along with the JSON file.



Assignment05.py | Enrollments.json | Mod05-Lab02-WorkingWithJSONFiles-NEW.py | MyLabData.json

```python
# ------------------------------------------------------------------------ #
# Title: Assignment05
# Desc: Using dictionaries, files, and exception handling.
# Change Log: (Who, When, What)
# GCarter, 11/11/2026, Created Script
# ------------------------------------------------------------------------ #

import json

# Define the Data Constants
MENU: str = '''
----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------
'''
FILE_NAME: str = 'Enrollments.json'

#Define the Data Variables
student_first_name: str = ''
student_last_name: str = ''
course_name: str = ''
student_data: dict = {}
students: list = []
file = None
menu_choice: str = ''

# When the program starts, read the file data into a list of dictionary rows
try:
    file = open(FILE_NAME, "r")
    students = json.load(file)
    file.close()
except FileNotFoundError as e:
    print("Text file must exist before running this script.\n")
    print("---Technical Error Message---")
    print(e, e.__doc__, type(e), sep='\n')
except Exception as e:
    print("There was a non-specific error!\n")
    print("---Technical Error Message---")
    print(e, e.__doc__, type(e), sep='\n')
finally:
    # Check if a file object exists and is still open
    if file is not None and file.closed == False:
        file.close()

# Present menu and process the data
while True:
    print(MENU)
    menu_choice = input("What would you like to do? ")
    print()
```

Assignment05.py | Enrollments.json

```python
    if menu_choice == "1":
        try:
            student_first_name = input("Enter student's first name: ")
            if not student_first_name.isalpha():
                raise ValueError("The first name should not contain numbers.")
            student_last_name = input("Enter student's last name: ")
            if not student_last_name.isalpha():
                raise ValueError("The last name should not contain numbers.")
            course_name = input("Please enter the name of the course: ")
            student_data = {'FirstName': student_first_name,
                            'LastName': student_last_name,
                            'CourseName': course_name}
            # Load it into our collection (list of lists)
            students.append(student_data)
            print(f"You have registered for {student_first_name} "
                  f"{student_last_name} for {course_name}.")
        except ValueError as e:
            print(e)
            print("---Technical Error Message---")
            print(e.__doc__)
            print(e.__str__())
        except Exception as e:
            print("There was a non-specific error!\n")
            print("---Technical Error Message---")
            print(e, e.__doc__, type(e), sep='\n')
        continue

    # show current data
    elif menu_choice == "2":
        print("-"*50)
        for student_data in students:
            print(f"Student {student_data['FirstName']} {student_data['LastName']} is "
                  f"enrolled in {student_data['CourseName']}.")
        print("-"*50)
        continue

    # save the data to the file
    elif menu_choice == "3":
        try:
            file = open(FILE_NAME, "w")
            json.dump(students, file, indent=2)
            file.close()
            print("The data has been saved in", FILE_NAME)
        except TypeError as e:
            print("Please check that the data is a valid JSON format.\n")
            print("---Technical Error Message---")
            print(e.__doc__, type(e), e.__str__(), sep='\n')
        except Exception as e:
            print("---Technical Error Message---")
            print("Built-In Python error info: ")
            print(e, e.__doc__, type(e), sep='\n')
        finally:
            if file is not None and file.closed == False:
                file.close()
        continue

    # stop the loop
    elif menu_choice == "4":
        break

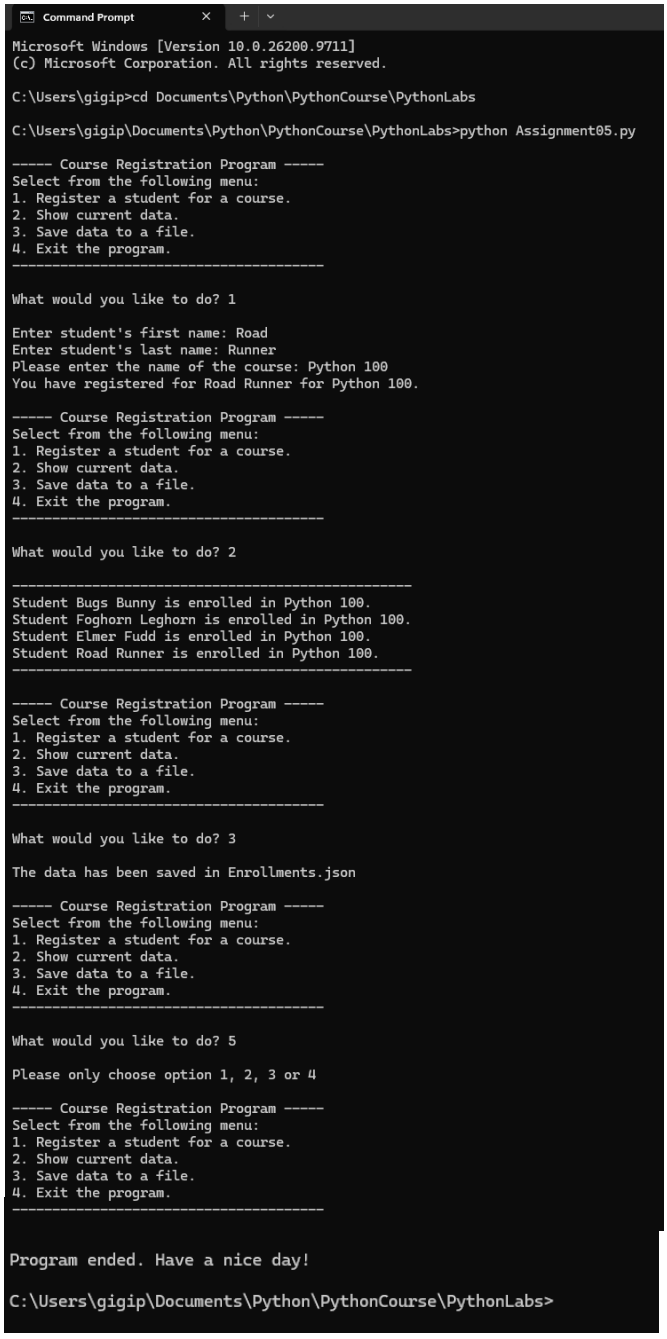
    else:
        print("Please only choose option 1, 2, 3 or 4")

print("Program ended. Have a nice day!")
```

Assignment05.py | Enrollments.json

```json
[
  {
    "FirstName": "Bugs",
    "LastName": "Bunny",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Foghorn",
    "LastName": "Leghorn",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Elmer",
    "LastName": "Fudd",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Road",
    "LastName": "Runner",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Micky",
    "LastName": "Mouse",
    "CourseName": "Python 100"
  }
]
```

After I ran the program in PyCharm, I checked it in the console:



```
Command Prompt
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\gigip>cd Documents\Python\PythonCourse\PythonLabs

C:\Users\gigip\Documents\Python\PythonCourse\PythonLabs>python Assignment05.py

----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------

What would you like to do? 1

Enter student's first name: Road
Enter student's last name: Runner
Please enter the name of the course: Python 100
You have registered for Road Runner for Python 100.

----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------

What would you like to do? 2

-------------------------------------------------
Student Bugs Bunny is enrolled in Python 100.
Student Foghorn Leghorn is enrolled in Python 100.
Student Elmer Fudd is enrolled in Python 100.
Student Road Runner is enrolled in Python 100.
-------------------------------------------------

----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------

What would you like to do? 3

The data has been saved in Enrollments.json

----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------

What would you like to do? 5

Please only choose option 1, 2, 3 or 4

----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------


Program ended. Have a nice day!

C:\Users\gigip\Documents\Python\PythonCourse\PythonLabs>
```



```
Command Prompt
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\gigip>cd Documents\Python\PythonCourse\A05

C:\Users\gigip\Documents\Python\PythonCourse\A05>python Assignment05.py

----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------

What would you like to do? 2

-------------------------------------------------
Student Bugs Bunny is enrolled in Python 100.
Student Foghorn Leghorn is enrolled in Python 100.
Student Elmer Fudd is enrolled in Python 100.
Student Road Runner is enrolled in Python 100.
Student Micky Mouse is enrolled in Python 100.
-------------------------------------------------

----- Course Registration Program -----
Select from the following menu:
1. Register a student for a course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
----------------------------------------

What would you like to do? 4

Program ended. Have a nice day!

C:\Users\gigip\Documents\Python\PythonCourse\A05>
```

Summary

In summary, Module 5 expands on Module 4's list of lists table to a list of dictionaries, where data values are stored under named keys instead of index numbers. It also introduced me to JSON (JavaScript Object Notation), compared it with CSV files, and showed me how the dump() and load() functions to write/save and read data. This module also showed me structured error handling using try and except, and how to catch exceptions and print custom messages for those errors. Finally, Module 5 introduced me to GitHub, a cloud-based platform for code sharing and collaboration. The demos, labs, live sessions, and this assignment were very helpful in growing my knowledge of Python.

Here is my GitHub repository link: https://github.com/geraldinepcarter/IntroToProg-Python-Mod05